Christian M. Lange

christianlange196@gmail.com Christian Lange
Google Scholar PhD Defense Full CV

Profile-Summary

- Experimental physicist specializing in quantum optics and nanophotonics, with a strong track record of rapidly building end-to-end systems. Led a research subgroup from an empty optical table to a *Nature Physics* publication in 2.5 years. Deep expertise in nanofabrication, tabletop optics, programming and automation, quantum theory, and materials synthesis.

Education

2026 – Present
- **Postdoctoral Researcher**, Purdue University, Quantum Photonic Integrated Design Center (QuPIDc),
  Advisor: Dr. Libai Huang

2021 – 2025
- **Ph.D. in Physics**, Purdue University
  Thesis: *Quantum Light–Matter Interactions with Organic Molecules*
  Advisor: Dr. Jonathan D. Hood

2017 – 2021
- **B.S. in Physics**, Utah State University
  Minors: Mathematics, Computer Science
  Summa Cum Laude GPA: 3.99 / 4.00
  Thesis: *The reflection of light from periodic conducting interfaces* (2021)
  Advisor: Dr. Tsung-Cheng Shen

Publications

- **Lange, C. M.**, E. Daggett, V. Walther, L. Huang, and J. D. Hood, "Superradiant and subradiant states in lifetime-limited organic molecules through laser-induced tuning", *Nature Physics*, vol. 20, pp. 836–842, 2024. DOI: 10.1038/s41567-024-02404-4.
- E. Daggett, **Lange, C. M.**, et al., "Many-body entanglement in solid-state emitters", *Nature Reviews Materials*, 2026. DOI: 10.1038/s41578-026-00893-8.
- **Lange, C. M.** et al., "A hybrid molecular–nanophotonic platform for on-chip cavity quantum electrodynamics and collective interactions", *ACS Nano*, 2026. DOI: 10.1021/acsnano.5c19465.
- **Lange, C. M.** and T.-C. Shen, "Fabrication of height-modulated carbon nanotube forests: Morphologies and prospects for broadband absorption", *Carbon Trends*, vol. 4, p. 100 070, 2021. DOI: 10.1016/j.cartre.2021.100070.
- A. D. Keni, **Lange, C. M.**, et al., "Vapor phase assembly of molecular emitter crystals for photonic integrated circuits", Under review; submitted to *ACS Photonics*. DOI: 10.48550/arXiv.2602.18517.
- E. Daggett, **Lange, C. M.**, et al., "Coherence in individual perovskite quantum dots", Maniscript in preparation.

Patents

- **Lange, C. M.** and J. D. Hood, *Photon emission with organic molecules and nanophotonics*, U.S. Provisional Patent Application, Purdue Research Foundation (PRF Ref: 2025-HOOD-71146-01), 2025.

Selected Projects

Nanofabrication

- **Si_3N_4 nanobeam cavities and grating couplers** — Designed/fabricated various cavity geometries with optical $Q \sim 10{,}000$ and mode volume $\sim (2.5\,\lambda/n)^3$ and grating couplers with 20% coupling efficiency.

Optics

- **Confocal microscopes** — Built several scanning confocal microscopes in different configurations (galvos vs. motorized stages and free-space vs. optical fiber spatial filtering).
- **Time-domain quantum emitter measurements** — Photon correlation Fourier spectroscopy, Hanbury Brown-Twiss measurement, and fluorescence lifetime of individual quantum emitters.
- **Lasers & cavities** — Aligned several MSquared Ti:sapphire lasers and high-finesse cavities.
- **Extreme cross-polarization extinction** — Developed precision optics for cross-polarization extinction with visibility > 70 dB.

Micromechanics

- **Crystal micromanipulator** — Positioned 30 μm crystals onto nanophotonic devices using a tapered fiber on a 3D micropositioner.
- **Motorized XY stage** — Built precision stage for heavy loads (> 25 lbs) with < 2 μm repeatability.

Programming, electronics, and automation

- **Experiment automation** — Python based GUI and custom drivers for remote/automated experiments: FAST Comtec and Swabian time taggers (Python), M Squared laser (TCP/IP), various optomechanical instruments (NI-DAQ, Python), and custom instruments (ARduino, RPi, ESP32).
- **Low-cost photon counter** — ESP32-based solution for low-time resolution photon counting.
- **Remote lab monitoring** — RPi/Arduino sensors + Grafana temperature monitoring.

Theory

- **Collective quantum systems** — QuTiP-based simulator for collective interactions.
- **Electromagnetic scattering** — Analytical and numerical modeling of reflection from periodic metal grids (undergraduate thesis topic).
- **Probability and linear algebra course development** — Assisted in developing courses for probability and statistics and linear algebra based on Jupyter notebooks.

Materials synthesis

- **DBT:anthracene crystal growth** — Built a custom two-stage tube furnace for vapor-phase growth of ultra-high aspect-ratio crystals and implemented a protocol for solution-based growth of DBT-doped anthracene nanocrystals.